1. Creating and Renaming Files/Directories

Create a directory named test_dir using mkdir.

Inside test_dir, create an empty file called example.txt.

Rename example.txt to renamed_example.txt using mv

output:

```
mohak@mohak-virtual-machine:~/Documents/Tutedude$ mkdir test_dir
mohak@mohak-virtual-machine:~/Documents/Tutedude$ cd test_dir/
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ touch exmaple.txt
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ ls
exmaple.txt
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ mv exmaple.txt renamed_example.txt
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ ls
renamed_example.txt
```

2. Viewing File Contents

Use cat to display the contents of /etc/passwd.

Display only the first 5 lines of /etc/passwd using head.

Display only the last 5 lines of /etc/passwd using tail.

output:

```
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111::/home/syslog:/usr/sbin/nologin
_apt:x:105:65534::/nonexistent:/usr/sbin/nologin
tss:x:106:113:TPM software stack,,,:/var/lib/tpm:/bin/false
uuidd:x:107:116::/run/uuidd:/usr/sbin/nologin
systemd-oom:x:108:117:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
tcpdump:x:109:118::/nonexistent:/usr/sbin/nologin
usbmux:x:111:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
```

```
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ man head
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ tail -n 5 /etc/passwd
jenkins:x:1001:1001::/home/jenkins:/bin/sh
dhcpcd:x:130:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
cups-browsed:x:131:122::/nonexistent:/usr/sbin/nologin
polkitd:x:999:999:User for polkitd:/:/usr/sbin/nologin
gnome-remote-desktop:x:996:996:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$
```

3.Searching for Patterns

Use grep to find all lines containing the word "root" in /etc/passwd.

```
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ grep -e "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
nm-openvpn:x:121:127:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$
```

4. Zipping and Unzipping

Compress the test_dir directory into a file named test_dir.zip using zip.

Unzip test_dir.zip into a new directory named unzipped_dir.

```
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ zip -r test_dir.zip test_dir
  adding: test_dir/ (stored 0%)
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ ls
renamed_example.txt  test_dir  test_dir.zip
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ unzip test_dir.zip
Archive:  test_dir.zip
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ ls
renamed_example.txt  test_dir  test_dir.zip
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$
```

5. Downloading Files

Use wget to download a file from a URL (e.g., https://example.com/sample.txt).

```
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ wget https://docs.google.com/spreadsheets/d/1eE-NhZQF
r545LkP4QNhTgXcZTtkMFeEPNyVXAflXia0/edit?gid=2073716385#gid=2073716385
--2026-02-20 21:09:10--  https://docs.google.com/spreadsheets/d/1eE-NhZQFr545LkP4QNhTgXcZTtkMFeEPNyVXAflXia0/edi
t?gid=2073716385
Resolving docs.google.com (docs.google.com)... 142.250.195.78, 2404:6800:4009:819::200e
Connecting to docs.google.com (docs.google.com)|142.250.195.78|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: ‘edit?gid=2073716385’

edit?gid=2073716385             [  <=>                                      ] 385.22K   972KB/s    in 0.4s

2026-02-20 21:09:18 (972 KB/s) - ‘edit?gid=2073716385’ saved [394465]

mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ ls
'edit?gid=2073716385'  'edit?tab=t.0'   renamed_example.txt   test_dir   test_dir.zip
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$
```

6. Changing Permissions

Create a file named secure.txt and change its permissions to read-only for everyone using chmod.

```
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ sudo chmod 777 renamed_example.txt
[sudo] password for mohak:
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ ls -lah
total 616K
drwxrwxr-x 3 mohak mohak 4.0K Feb 20 21:09  .
drwxrwxr-x 3 mohak mohak 4.0K Feb 20 20:57  ..
-rw-rw-r-- 1 mohak mohak 386K Feb 20 21:09 'edit?gid=2073716385'
-rw-rw-r-- 1 mohak mohak 211K Feb 20 21:08 'edit?tab=t.0'
-rwxrwxrwx 1 mohak mohak    0 Feb 20 20:57  renamed_example.txt
drwxrwxr-x 2 mohak mohak 4.0K Feb 20 21:03  test_dir
-rw-rw-r-- 1 mohak mohak  168 Feb 20 21:06  test_dir.zip
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$
```

7. Working with Environment Variables

Use export to set a new environment variable called MY_VAR with the value "Hello, Linux!".

```
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ export MY_VAR="Hello, Linux!"
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$ echo $MY_VAR
Hello, Linux!
mohak@mohak-virtual-machine:~/Documents/Tutedude/test_dir$
```